Žan Grad – Curriculum Vitae

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Basic information

- Nationality: Slovenian
- Place and date of birth: Celje (Slovenia), October 6, 1993
- Current institution: KU Leuven, Belgium

Education

Doctoral degree in Mathematics (Ph. D.) — Graduated June 27, 2025

Instituto Superior Técnico, University of Lisbon

- Member of Centre for Mathematical Analysis, Geometry, and Dynamical Systems
- Topic: *Fundamentals of Lie categories and Yang–Mills theory for multiplicative Ehresmann connections*.
- Thesis awarded with honors and distinction (*summa cum laude*).
- Advisers: Ioan Mărcuț and Pedro Resende.

Master's degree in Mathematics (M. Sc.) — Graduated September 10, 2020

University of Ljubljana, Faculty of Mathematics and Physics

- Topic: *Gauge field theory, Yang–Mills–Higgs equations and spin structures on Lorentzian manifolds*.
- Thesis awarded with the Prešeren award of University of Ljubljana.
- Adviser: Sašo Strle.

Bachelor's degree in Physics (B. Sc.) — Graduated July 7, 2017

University of Ljubljana, Faculty of Mathematics and Physics

- Topic: *Wave function collapse upon measuring the number of photons in a cavity.*
- Adviser: Tomaž Rejec.

Employment

Postdoctoral researcher in Pure Mathematics — Sep 2025 – Ongoing

KU Leuven, Belgium

- Field of research: Poisson geometry and Lie theory.
- Adviser: Marco Zambon.

Assistant professor of mathematics — Sep 2020 – Sep 2021

University of Ljubljana, Faculty of Mathematics and Physics

- Employed full time for two semesters, with an average load of 10 hours of teaching per week.
- Held exercise sessions, constructed written exams, graded students.
- Taught two master-level courses: Analysis on manifolds and Differential geometry.
- Taught five bachelor-level courses: Analysis IV for financial mathematicians, Mathematics 2 for physicists, Mathematics 4 for physicists, Basics of mathematical analysis for computer engineers, Mathematics for pharmacists.

Recorded exercise sessions for these courses are available on my personal webpage, in the teaching section.

Publications

[1] Žan Grad, *Fundamentals of Lie categories*, J. Noncommut. Geom. **19** (2025), no. 1, 211—248, DOI 10.4171/JNCG/563. (arXiv).

[2] ———, *Covariant derivatives in the representation-valued Bott–Shulman–Stasheff and Weil complex.* (arXiv).

[3] ———, *Yang–Mills theory for multiplicative Ehresmann connections.* In preparation.

Talks

- *Yang-Mills theory for Lie algebroids*, Workshop "Graded geometry of gauge theories", University of Mons, March 12, 2026.
- *On obstructions to the existence of Cartan connections*, Poisson seminar, KU Leuven, February 25, 2026.
- *Lie groupoids and algebroids*, Junior Methusalem Seminar, KU Leuven, December 11, 2025.
- *Multiplicative Ehresmann connections*, Analysis & Geometry Seminar, University of Antwerp, November 19, 2025.
- *Infinitesimal multiplicative forms*, Poisson seminar, KU Leuven, October 21, 2025.
- *Multiplicative Ehresmann connections*, Poisson seminar, KU Leuven, September 30, 2025.
- *Yang–Mills theory for multiplicative Ehresmann connections*, Higher Geometric Structures along the Lower Rhine XIX, University of Cologne, September 25-26 2025.
- *Yang–Mills theory for multiplicative Ehresmann connections*, University of Illinois Urbana–Champaign (Symplectic and Poisson Geometry Seminar), April 28, 2025.
- *Double complexes of representation-valued forms*, Universität zu Köln, Poisson Geometry Seminar, February 5, 2025.
- *Double complexes of representation-valued forms*, JMU Würzburg, Oberseminar Deformationsquantisierung und Geometrie, January 31, 2025.
- *The exterior covariant derivative of multiplicative forms*, IMPA, Rio de Janeiro, November 14, 2024.
- *Crash course on Symplectic fibrations*, Radboud University Nijmegen, April 2024.
- *Mini course on Lie categories*, Lie groupoid and algebroid week in Coimbra II, July 2023.
- *Lie categories*, Geometry Seminar of Lisbon, Instituto Superior Técnico, Universidade de Lisboa, May 9, 2023.
- *Lie categories*, PhD seminar of Radboud University Nijmegen, April 2023.
- Two talks on *Lie categories*, Topology Seminar of Faculty of Mathematics and Physics, University of Ljubljana, March 20 and March 27, 2023.
- A series of lectures on Lie groupoids, Instituto Superior Técnico, Universidade de Lisboa, Spring semester 2021/22.
- *Mini course on Minkowski space and the theory of relativity*, Summer school of Mathematics and Physics at University of Ljubljana, August 2019.

In the academic year 2025/26, I am organizing the weekly Poisson seminar at KU Leuven. In the spring semester of 2024, I organized an online seminar on the integration of Lie algebroids to Lie groupoids, where we discussed and proved the well-known result by Crainic and Fernandes stating the precise obstructions to the integrability of Lie algebroids. This seminar was co-organized by fellow PhD students from University of Coimbra and Radboud University.

Conferences and Workshops

The following list contains attended and planned conferences and workshops. The ones colored in red are those on which I am a part of the organizing committee.

- (Planned) Poisson School & Conference 2026, August 3-14, 2026 (Antwerp & Leuven)
- (Planned) Field theory and higher geometric structures, Poisson Intensive Period, June 29-July 3, 2026 (Leuven)
- (Planned) Lie groupoid and Lie algebroid week IV, June 22–26, 2026 (Coimbra)
- (Planned) Deformation problems in and around Poisson geometry, Poisson Intensive Period, June 1-5, 2026 (Leuven)
- (Planned) Gone fishing 2026, April 23-26, 2026 (Montreal)
- Higher Geometric Structures along the Lower Rhine XX, April 16-17, 2026 (Utrecht).
- Fifty years of BRST, March 23-27, 2026 (Munich)
- Higher Geometric Structures along the Lower Rhine XIX, September 25-26 2025 (Cologne).
- Interactions of Poisson Geometry, Lie Theory and Symmetry, June 30-July 4, 2025 (Lisbon).
- Higher Geometric Structures along the Lower Rhine XVIII, January 23-24, 2025 (Nijmegen).
- XI Workshop on Poisson Geometry and Related Topics, November 4-8, 2024 (Manaus).
- XXXII International Fall Workshop on Geometry and Physics, September 2-5, 2024 (Coimbra).
- Poisson School & Conference 2024, July 1-12, 2024 (Naples).

- Lie groupoid and algebroid week III, June 3-7, 2024 (Coimbra).
- Higher Geometric Structures along the Lower Rhine XVII, May 2-3, 2024 (Bonn).
- Lie groupoid and algebroid week II, July 10-14, 2023 (Coimbra).
- The many interactions of symplectic and Poisson geometry, June 19-23, 2023 (Paris).
- Lie groupoid and algebroid week I, September 26-30, 2022 (Coimbra).
- Poisson School & Conference 2022, July 18-29, 2022 (attended online).

Posters

- The exterior covariant derivative of multiplicative forms for XI Workshop on Poisson Geometry and Related Topics (Poisson in the Amazon), November 2024.

Grants

- COST CaLISTA European grant for short-term scientific missions. This grant enabled me to visit Ioan Mărcuț in person in Nijmegen, in the period of April and May 2024.
- PhD grant UI/BD/152069/2021 by Fundação para a Ciência e a Tecnologia (FCT), Portugal.

Awards

- *Prešeren award of Faculty of Mathematics and Physics, University of Ljubljana*, for master's thesis *Gauge field theory, Yang–Mills–Higgs equations and spin structures on Lorentzian manifolds*, 2020.

Mentoring

- Master's student: *Matthijs Lau*, Radboud University Nijmegen. Topic: *Symplectic groupoid fibrations.* Graduated in June 2025. This is a joint supervision with Ioan Mărcuț.

Other

Physics

I am interested in applying modern mathematical methods to describe the theory of general relativity, quantum mechanics, many-body quantum mechanics, quantum field theory, classical mechanics, classical field theory (hydrodynamics, electromagnetism, elastomechanics and gauge field theory).

Languages

Native speaker of Slovenian language. I am fluent in both written and spoken English (C2 proficiency).

Programming languages

I have advanced knowledge of languages of the web: Python, PHP, Javascript, HTML, CSS. Principles of backend and frontend frameworks, together with concrete advanced implementations of frameworks Laravel (PHP) in VueJS (Javscript). To support myself, I worked a part-time job as a full-stack web developer between 2010 and 2016. I also have basic knowledge of Mathematica and some familiarity with C.